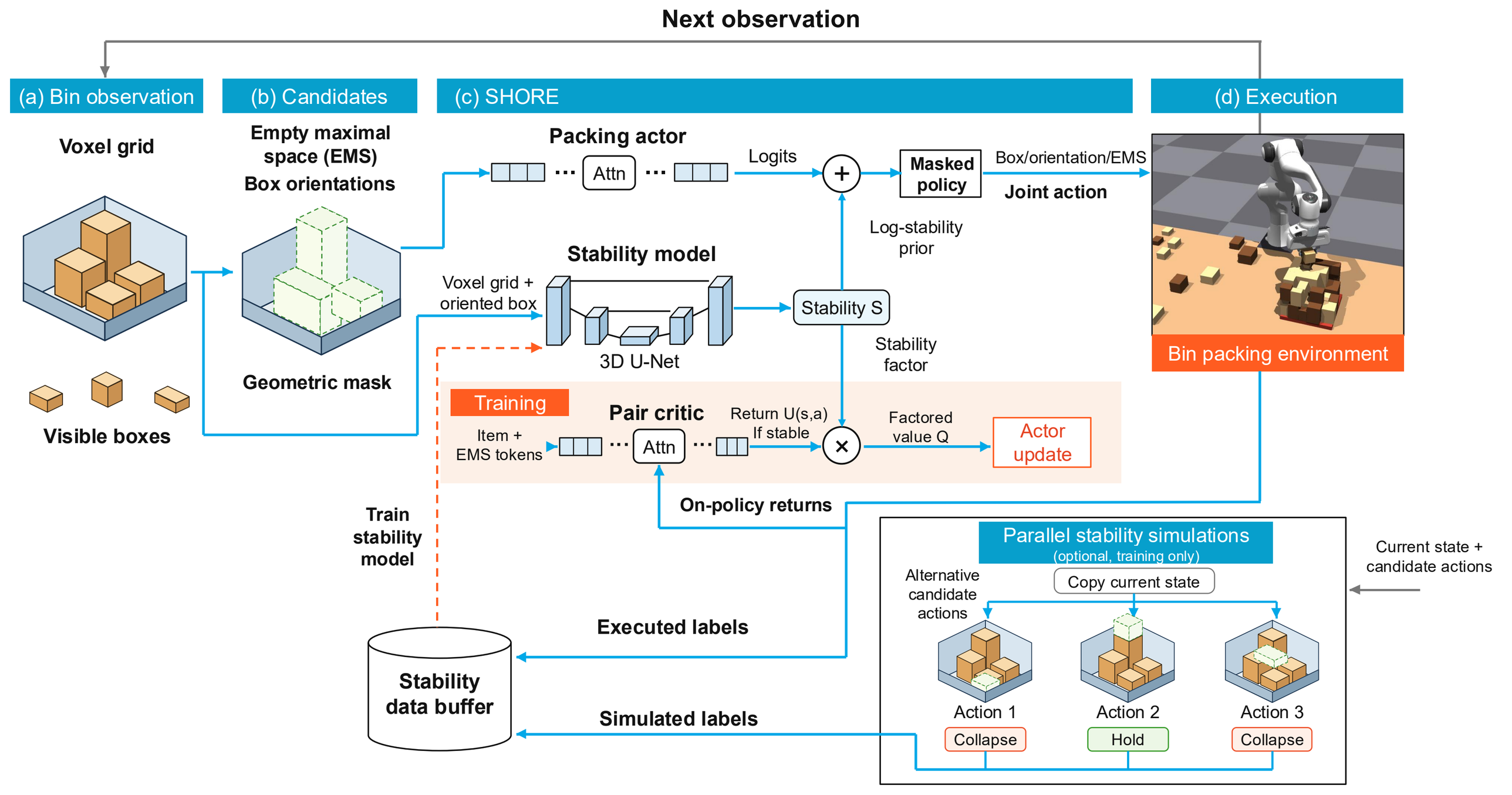

Next observation
(a) Bin observation
(b) Candidates
(c) SHORE
(d) Execution
Voxel grid
Empty maximal space (EMS)
Box orientations
Geometric mask
Visible boxes
Packing actor
Attn
Logits
Masked policy
Box/orientation/EMS
Joint action
Log-stability prior
Stability model
Voxel grid + oriented box
3D U-Net
Stability S
Stability factor
Bin packing environment
Training
Pair critic
Item + EMS tokens
Attn
Return U(s,a) If stable
Factored value Q
Actor update
On-policy returns
Train stability model
Executed labels
Stability data buffer
Simulated labels
Parallel stability simulations
(optional, training only)
Alternative candidate actions
Copy current state
Action 1
Collapse
Action 2
Hold
Action 3
Collapse
Current state + candidate actions